

Non-Graded Project: Designing an Autonomous E-commerce Support Crew

In this project, you will step into the role of an AI Systems Architect to design, build, and document a multi-agent AI system using Flowise. You will create a "crew" of specialized AI agents to handle various customer support queries for a growing e-commerce store, orchestrating their collaboration to ensure efficient and responsible operation.

Learning Objectives

- Design a multi-agent system architecture with at least three specialized agent roles and a central orchestrator.
- Implement a coordinated workflow in Flowise to route user queries to the appropriate agent.
- Develop a basic governance plan that identifies potential risks and proposes mitigation strategies.
- Analyze the trade-offs between agent autonomy and system-level control in a real-world scenario.



Assignment Components

Introductory Slide:

- Title: Project Command Center: Deploying Your E-commerce AI Crew
- Subtitle: Architect, build, and govern an intelligent multi-agent system to automate online customer support.

Tools to be used:

- Flowise AI (web-based or locally installed version)
- A text editor (e.g., Google Docs, MS Word, Notion) for documentation.

Scenario

You are the lead AI Systems Architect for "Future Gadgets Inc.," a popular online store experiencing rapid growth. The human customer support team is overwhelmed with a high volume of repetitive queries falling into three main categories:

1. Product Information: General questions about product features, specifications, and availability.
2. Order Status: Specific inquiries about where an order is and its estimated delivery date.

3. Promotions & Deals: Questions about current sales, discounts, and special offers.

Your mission is to design and build a prototype of a multi-agent AI system in Flowise that can autonomously handle these queries. The system must be able to understand the user's intent and route the query to the correct specialist agent. Crucially, management is concerned about security and reliability, so your design must also include a governance plan that addresses potential risks.

Instructions for learners

Part 1: System Design & Specialist Agent Creation

1 Plan your architecture. Before building in Flowise, sketch a simple diagram of your system. You will need one "Orchestrator Agent" and three "Specialist Agents." Decide the name and specific function for each specialist.

2 Create the "Product Info Agent." In Flowise, create a new chatflow. Design this agent to answer general questions about "Future Gadgets Inc." products.
Tip: You can use a simple LLM Chain with a prompt that gives it the persona of a product expert. For a more advanced approach, you could use a Document Loader with a text file containing product FAQs.

3

Create the "Order Status Agent." Create a second chatflow. This agent's only job is to respond to questions about order status. Since it can't access a real database, configure its prompt to explain its function and ask the user for an order number, after which it can provide a simulated static response (e.g., "Orders are typically delivered within 3-5 business days.").

4

Create the "Promotions Agent." Create a third chatflow. Design this agent to inform users about the latest deals. Configure its prompt to act as a sales assistant and provide information on a fictional sale (e.g., "We are currently offering a 20% discount on all smartwatches!").

5

Save each chatflow with a clear name (e.g., "Product Agent," "Order Agent").

Part 2: Building the Central Orchestrator

1

Create the main "Orchestrator Agent." Create a new, primary chatflow. This will be the user's single point of contact.

2

Add a Router Chain. From the "Chains" menu in Flowise, add the Multi Prompt Chain or a similar routing tool. This node will serve as the brain of your orchestrator.

Define the routing logic. Configure the router to analyze the user's input and decide which specialist agent to forward the query to. Create descriptions for each route that will help the router make its decision.

3

For example:

Name: product_info | Description: "Routes questions about product features, details, or specifications."

Name: order_status | Description: "Routes questions about order tracking, shipping, or delivery dates."

4

Connect the Specialist Agents. Use the "Chatflow" node in Flowise to link each route in your orchestrator to the corresponding specialist agent chatflow you built in Part 1. Your final canvas should show the orchestrator connected to the three specialist agents.

5 Test your system. Use the chat interface in the orchestrator chatflow to ask different types of questions and verify that the query is being routed to the correct specialist agent.

Part 3: Governance & Documentation

1 Document your work. In a separate text document, complete the deliverables outlined below. This includes explaining your design, analyzing risks, and reflecting on the process.

Deliverables for Submission (AI Grading)

1 Deliverable 1: Final System Architecture

Question Prompt: "Upload a single, full-screen screenshot of your final 'Orchestrator Agent' canvas in Flowise. Your screenshot must clearly show the central routing node and its connections to the at least three specialist agent chatflows you designed."

Response Type: File Upload

File Types: PNG, JPG, JPEG

2 Deliverable 2: Design Methodology Explanation

Question Prompt: "Submit a brief document (150-200 words) explaining your design methodology. In your explanation, describe the specific purpose of each of your three specialist agents, and justify the routing descriptions you created for your orchestrator agent, explaining how they help the system make accurate decisions."

Response Type: Rich Text or File Upload

File Types: PDF, DOCX, TXT

3 Deliverable 3: Risk & Governance Analysis

Question Prompt: "In a short analysis (200-300 words), identify at least two distinct potential risks for your e-commerce multi-agent system (e.g., agent providing incorrect information, handling sensitive user data, agent 'hallucinations'). For each risk, propose one specific governance control or mitigation strategy you would implement to ensure safe and compliant operation."

Response Type: Rich Text, txt, pdf, docx

4 Deliverable 4: Reflection & Learning Insights

→ Question Prompt: "How did you approach this AI agents design and what key insights will you apply in future projects? (250 words)"

→ Response Type: File Upload

→ File Types: Rich Text, txt, pdf, docx

Share The Project

Document: Save the report in PDF format.

Upload: Share in the “Peer Review” area of the course shell with a brief description.

Instructions for Documenting and Sharing The Project for Peer Review

1. Document the Project

- Use word processing software to create your report.
- Ensure all sections of the project document structure are completed.
- Proofread and edit your report for clarity and accuracy.

2. Share the Project

- Save your report in PDF format.
- Upload your documents to the “Peer Review” area in the course shell.
- Provide a brief description of your project in the submission post.

Instructions for Documenting and Sharing The Project for Peer Review

3. Peer Review

- Review the projects submitted by your peers.
- Provide constructive feedback and comments on their work.